

Programming for Data Science

Medical Insurance Cost Analysis

Data analysis using Pandas, NumPy and Matplotlib

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Subject	Programming for Data Science
Dataset	Medical Cost Personal Datasets (Kaggle)
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Chapter 1: Introduction and Data Description

1.1 Introduction

Health insurers must price policies so that premiums cover the expected medical costs of each beneficiary. Understanding which personal attributes drive those costs helps an insurer set fair premiums, design wellness programmes and manage risk. This project analyses individual medical insurance charges billed in the United States and asks: *which factors (age, sex, BMI, number of children, smoking status, region) explain the variation in charges, and by how much?*

The analysis is carried out in Python with three core libraries: **Pandas** for loading, cleaning, grouping and reshaping the data; **NumPy** for vectorised statistics, correlation and a least-squares regression; and **Matplotlib** for visual storytelling.

1.2 Data source

Dataset: Medical Cost Personal Datasets, published on Kaggle by Miri Choi (<https://www.kaggle.com/datasets/mirichoi0218/insurance>). The data originates from the book *Machine Learning with R* by Brett Lantz and is simulated from U.S. Census Bureau demographic statistics. It is a secondary data source in CSV format (insurance.csv).

1.3 Data description

The file has **1,338 rows** (one per insured person) and **7 columns**: 3 categorical and 4 numerical variables, which meets the requirement of at least 510 rows with both variable types.

Variable	Type	Description
age	Numerical (int)	Age of the primary beneficiary (18-64 years)
sex	Categorical	Gender of the policy holder (male / female)
bmi	Numerical (float)	Body mass index, kg/m ² (ideal 18.5-24.9)
children	Numerical (int)	Number of children / dependents covered (0-5)
smoker	Categorical	Whether the beneficiary smokes (yes / no)
region	Categorical	U.S. residential area: northeast, northwest, southeast, southwest
charges	Numerical (float)	Individual medical costs billed by health insurance (USD) - target variable

1.4 Objectives

- Profile and clean the dataset (types, missing values, duplicates).
- Compare charges across smoker, sex, region, age and BMI groups.
- Quantify the distribution, skewness and outliers of charges.
- Measure correlation of each factor with charges and fit a linear model.
- Visualise the key relationships and derive business recommendations.

Chapter 2: Data Analysis using Python

All code is in `insurance_analysis.py`. Each application below shows the code, the output produced when it runs, and the inference drawn.

Part 2.1: Pandas

Pandas Application 1 - Loading, profiling and cleaning

Objective: Read the CSV, check data types, missing values and duplicates, convert text columns to categories and summarise the numbers.

Code

```
1 def pandas_1_load_and_profile():
2     """Load the Kaggle CSV, profile its structure and clean it."""
3     df = pd.read_csv(DATA_FILE)
4     print("Shape (rows, columns):", df.shape)
5     print("Data types:", dict(df.dtypes.astype(str)))
6     print("Missing values (total) :", df.isna().sum().sum())
7     print("Duplicate rows found :", df.duplicated().sum())
8     df = df.drop_duplicates().reset_index(drop=True)
9     print("Shape after cleaning :", df.shape)
10
11     for col in ["sex", "smoker", "region"]:
12         df[col] = df[col].astype("category")
13         share = df[col].value_counts(normalize=True).mul(100).round(1)
14         print(f"{col:<7} distribution (%)", share.to_dict())
15
16     print("\nSummary statistics of numerical variables:")
17     print(df.describe().round(2))
18     return df
```

Result

```
Shape (rows, columns): (1338, 7)
Data types: {'age': 'int64', 'sex': 'str', 'bmi': 'float64', 'children': 'int64', 'smoker': 'str', 'region': 'str', 'charges': 'float64'}
Missing values (total) : 0
Duplicate rows found : 1
Shape after cleaning : (1337, 7)
sex distribution (%): {'male': 50.5, 'female': 49.5}
smoker distribution (%): {'no': 79.5, 'yes': 20.5}
region distribution (%): {'southeast': 27.2, 'southwest': 24.3, 'northeast': 24.2, 'northwest': 24.2}

Summary statistics of numerical variables:
   count  age      bmi  children  charges
count  1337.00  1337.00  1337.00  1337.00
mean    39.22   30.66    1.10  13279.12
std     14.04    6.10    1.21  12110.36
min     18.00   15.96    0.00   1121.87
25%     27.00   26.29    0.00   4746.34
50%     39.00   30.40    1.00   9386.16
75%     51.00   34.70    2.00  16657.72
max     64.00   53.13    5.00  63770.43
```

Inference

The dataset is clean: no missing values and only **one duplicate row**, which was removed, leaving 1,337 records. The sample is balanced by sex (50.5% male) and region (~24-27% each), but only **20.5% are smokers**. Charges range from \$1,122 to \$63,770; the mean (\$13,279) is far above the median (\$9,386), a first sign that a small group of very expensive customers pulls the average up. The mean BMI of 30.7 means the average beneficiary is clinically obese.

Pandas Application 2 - Group comparisons with groupby and pivot_table

Objective: Compare count, mean, median and maximum charges by smoker, sex and region, and cross smoker with region in a pivot table.

Code

```
1 def pandas_2_group_comparisons(df):
2     """Compare insurance charges across smoker, sex and region groups."""
3     for col in ["smoker", "sex", "region"]:
4         summary = (df.groupby(col, observed=True)["charges"]
5                     .agg(["count", "mean", "median", "max"])
6                     .sort_values("mean", ascending=False)
7                     .round(0))
8         print(f"\n--- Charges by {col} ---")
9         print(summary)
10
11     pivot = pd.pivot_table(df, values="charges", index="region",
12                             columns="smoker", aggfunc="mean", observed=True)
13     pivot["smoker / non-smoker"] = pivot["yes"] / pivot["no"]
14     print("\nMean charges: region x smoker")
15     print(pivot.round(1))
```

Result

```
--- Charges by smoker ---
      count    mean  median    max
smoker
yes       274  32050.0  34456.0  63770.0
no       1063   8441.0   7346.0  36911.0

--- Charges by sex ---
      count    mean  median    max
sex
male       675  13975.0  9378.0  62593.0
female     662  12570.0  9413.0  63770.0

--- Charges by region ---
      count    mean  median    max
region
southeast   364  14735.0  9294.0  63770.0
northeast   324  13406.0  10058.0  58571.0
northwest   324  12451.0   8977.0  60021.0
southwest   325  12347.0   8799.0  52591.0

Mean charges: region x smoker
smoker      no      yes  smoker / non-smoker
region
northeast   9165.5  29673.5                3.2
northwest   8582.5  30192.0                3.5
southeast   8032.2  34845.0                4.3
southwest   8019.3  32269.1                4.0
```

Inference

Smoking is the dominant difference: smokers are charged **\$32,050 on average vs \$8,441** for non-smokers, about **3.8 times more**. Sex makes little difference (medians are almost identical: \$9,378 vs \$9,413). The southeast has the highest mean charge (\$14,735), and the pivot table shows this is where the smoker premium is largest (4.3x), so regional differences are mainly about smoking and BMI, not the region itself.

Pandas Application 3 - Feature engineering with pd.cut

Objective: Group ages into bands and BMI into WHO categories, then cross-tabulate mean charges.

Code

```
1 def pandas_3_feature_engineering(df):
2     """Create age groups and WHO BMI categories, then cross-tabulate."""
3     df["age_group"] = pd.cut(df["age"], bins=[17, 29, 39, 49, 64],
4                             labels=["18-29", "30-39", "40-49", "50-64"])
5     df["bmi_class"] = pd.cut(df["bmi"], bins=[0, 18.5, 25, 30, 100],
6                             labels=["Underweight", "Normal", "Overweight", "Obese"],
7                             right=False)
8
9     print("Customers per BMI class:")
10    print(df["bmi_class"].value_counts().sort_index().to_string())
11
12    table = pd.pivot_table(df, values="charges", index="age_group",
13                          columns="smoker", aggfunc="mean", observed=True).round(0)
14    print("\nMean charges by age group and smoking status:")
15    print(table)
16
17    obese_smoker = (df.assign(obese=df["bmi_class"] == "Obese")
18                  .groupby(["smoker", "obese"], observed=True)["charges"]
19                  .mean().unstack().round(0))
20    print("\nMean charges: smoker x obese (BMI >= 30)")
21    print(obese_smoker)
22    return df
```

Result

```
Customers per BMI class:
bmi_class
Underweight    20
Normal         225
Overweight     386
Obese          706

Mean charges by age group and smoking status:
smoker      no      yes
age_group
18-29      4427.0  27518.0
30-39      6337.0  30271.0
40-49      9183.0  32655.0
50-64     13431.0  38748.0

Mean charges: smoker x obese (BMI >= 30)
obese      False    True
smoker
no         7977.0   8856.0
yes        21363.0  41558.0
```

Inference

Over half of beneficiaries (706 of 1,337) are **obese** and only 225 have a normal BMI. Charges rise steadily with age in both groups, e.g. non-smokers go from \$4,427 (18-29) to \$13,431 (50-64). The key finding is an **interaction**: obesity adds only about \$900 for non-smokers, but for smokers it almost **doubles** the cost (\$21,363 to \$41,558). Obese smokers are the highest-risk segment.

Part 2.2: NumPy

NumPy Application 1 - Distribution, outliers and skewness

Objective: Use NumPy to compute central tendency, spread, IQR outliers and skewness, and test a log transform.

Code

```
1 def numpy_1_distribution_statistics(df):
2     """Describe the charges distribution and fix its skew with a log transform."""
3     charges = df["charges"].to_numpy()
4
5     def skewness(x):
6         return np.mean((x - x.mean()) ** 3) / np.std(x) ** 3
7
8     q1, med, q3 = np.percentile(charges, [25, 50, 75])
9     iqr = q3 - q1
10    upper = q3 + 1.5 * iqr
11    print(f"Mean = {np.mean(charges):10.2f}   Median = {med:10.2f}")
12    print(f"Std   = {np.std(charges, ddof=1):10.2f}   IQR   = {iqr:10.2f}")
13    print(f"Upper outlier fence (Q3 + 1.5*IQR) = {upper:.2f}")
14    outliers = charges[charges > upper]
15    print(f"Outliers: {outliers.size} ({outliers.size / charges.size:.1%} of rows)")
16
17    smoker = df["smoker"].to_numpy() == "yes"
18    print(f"Share of outliers who smoke: {np.mean(smoker[outliers > upper]):.1%}")
19
20    log_charges = np.log(charges)
21    print(f"\nSkewness of charges      : {skewness(charges):.3f}")
22    print(f"Skewness of log(charges): {skewness(log_charges):.3f}")
```

Result

```
Mean   = 13279.12   Median = 9386.16
Std    = 12110.36   IQR    = 11911.37
Upper outlier fence (Q3 + 1.5*IQR) = 34524.78
Outliers: 139 (10.4% of rows)
Share of outliers who smoke: 97.8%

Skewness of charges      : 1.514
Skewness of log(charges): -0.090
```

Inference

Charges are strongly **right-skewed** (skewness 1.51). 139 beneficiaries (10.4%) lie above the outlier fence of \$34,525, and **97.8% of these outliers are smokers**, so the outliers are real high-risk customers, not data errors, and should be kept. Taking the log reduces skewness to -0.09 (nearly symmetric), which is useful if the data is later used in statistical models.

NumPy Application 2 - Correlation matrix with np.corrcoef

Objective: Encode smoker as 0/1, build a feature matrix with np.column_stack and measure Pearson correlation with charges.

Code

```
1 def numpy_2_correlation(df):
2     """Pearson correlation matrix with np.corrcoef (smoker encoded as 0/1)."""
3     cols = ["age", "bmi", "children", "smoker", "charges"]
4     X = np.column_stack([
5         df["age"].to_numpy(float),
6         df["bmi"].to_numpy(float),
7         df["children"].to_numpy(float),
8         np.where(df["smoker"].to_numpy() == "yes", 1.0, 0.0),
9         df["charges"].to_numpy(float),
10    ])
11    corr = np.corrcoef(X, rowvar=False)
12    print("Correlation matrix:")
13    print(pd.DataFrame(corr, index=cols, columns=cols).round(3))
14
15    with_charges = corr[:, -1]
16    print("\nVariables ranked by |r| with charges:")
17    for i in np.argsort(-np.abs(with_charges)):
18        print(f"    {cols[i]:<9} r = {with_charges[i]:+.3f}")
19    return cols, corr
```

Result

```
Correlation matrix:
      age    bmi  children  smoker  charges
age    1.000  0.109   0.042  -0.026   0.298
bmi    0.109  1.000   0.013   0.004   0.198
children 0.042  0.013   1.000   0.007   0.067
smoker  -0.026  0.004   0.007   1.000   0.787
charges  0.298  0.198   0.067   0.787   1.000

Variables ranked by |r| with charges:
smoker    r = +0.787
age        r = +0.298
bmi        r = +0.198
children   r = +0.067
```

Inference

Smoking has a **strong positive correlation ($r = 0.79$)** with charges, far above age (0.30) and BMI (0.20). Number of children has almost no linear link (0.07). The predictors are nearly uncorrelated with each other (all $|r| < 0.11$), so each one contributes separate information and there is no multicollinearity problem for a regression model.

NumPy Application 3 - Multiple linear regression with `np.linalg.lstsq`

Objective: Estimate how much each factor adds to charges using ordinary least squares solved with linear algebra, and make a prediction.

Code

```
1 def numpy_3_linear_regression(df):
2     """Fit charges = b0 + b1*age + b2*bmi + b3*children + b4*smoker by least squares."""
3     X = np.column_stack([
4         np.ones(len(df)),
5         df["age"].to_numpy(float),
6         df["bmi"].to_numpy(float),
7         df["children"].to_numpy(float),
8         (df["smoker"].to_numpy() == "yes").astype(float),
9     ])
10    y = df["charges"].to_numpy(float)
11
12    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
13    y_hat = X @ beta
14    r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum((y - y.mean()) ** 2)
15    rmse = np.sqrt(np.mean((y - y_hat) ** 2))
16
17    names = ["intercept", "age", "bmi", "children", "smoker"]
18    print("Estimated coefficients:")
19    for n, b in zip(names, beta):
20        print(f"    {n:<10} {b:12.2f}")
21    print(f"\nR-squared = {r2:.3f}    RMSE = {rmse:.2f}")
22
23    profile = np.array([1, 40, 30, 2, 0])
24    print(f"Predicted charge, 40y, BMI 30, 2 kids, non-smoker: {profile @ beta:9.2f}")
25    profile[-1] = 1
26    print(f"Predicted charge, same person but a smoker      : {profile @ beta:9.2f}")
```

Result

```
Estimated coefficients:
intercept      -12098.82
age             257.77
bmi             321.87
children        472.98
smoker          23810.40

R-squared = 0.750    RMSE = 6058.64
Predicted charge, 40y, BMI 30, 2 kids, non-smoker:   8814.16
Predicted charge, same person but a smoker           : 32624.56
```

Inference

Four simple variables explain **75% of the variation** in charges ($R^2 = 0.750$). Holding other factors equal, each extra year of age adds about **\$258**, each BMI point about **\$322**, each child about \$473, and being a smoker adds about **\$23,810**. A 40-year-old with BMI 30 and two children is predicted at \$8,814 as a non-smoker but \$32,625 as a smoker. The RMSE of about \$6,059 shows the model misses the smoker-obesity interaction found in Pandas Application 3.

Part 2.3: Matplotlib

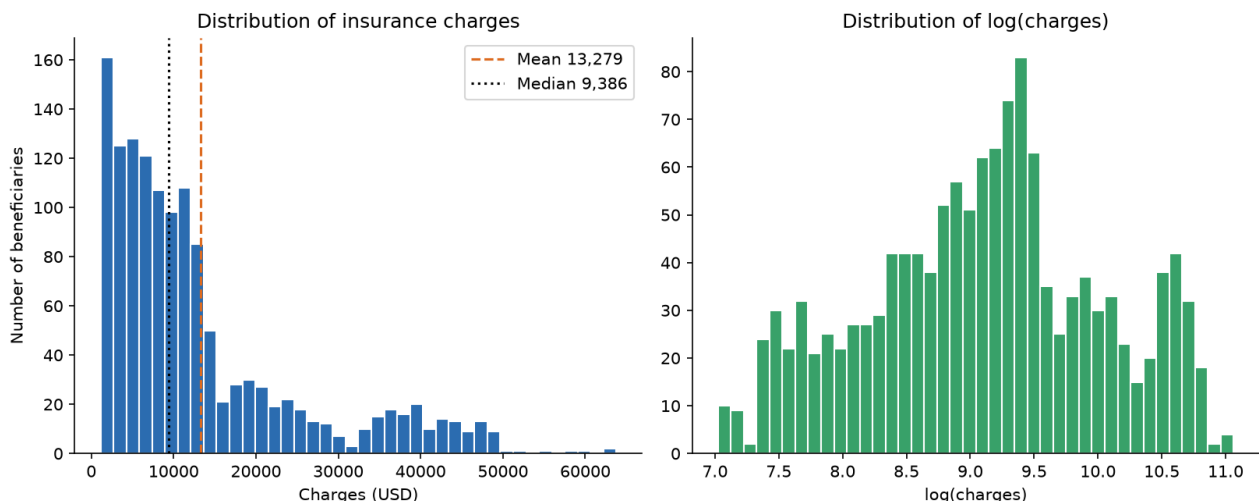
Matplotlib Application 1 - Histograms of charges

Objective: Visualise the shape of the charges distribution before and after a log transformation.

Code

```
1 def matplotlib_1_distribution(df):
2     """Histogram of charges (raw and log scale) with mean/median lines."""
3     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4.5))
4     charges = df["charges"]
5
6     ax1.hist(charges, bins=40, color="blue", edgecolor="white")
7     ax1.axvline(charges.mean(), color="red", ls="--",
8                 label=f"Mean {charges.mean():,.0f}")
9     ax1.axvline(charges.median(), color="black", ls=":",
10                label=f"Median {charges.median():,.0f}")
11     ax1.set_title("Distribution of insurance charges")
12     ax1.set_xlabel("Charges (USD)")
13     ax1.set_ylabel("Number of beneficiaries")
14     ax1.legend()
15
16     ax2.hist(np.log(charges), bins=40, color="green", edgecolor="white")
17     ax2.set_title("Distribution of log(charges)")
18     ax2.set_xlabel("log(charges)")
19     for ax in (ax1, ax2):
20         ax.spines[["top", "right"]].set_visible(False)
21
22     fig.tight_layout()
23     fig.savefig(FIG_DIR / "mpl_1_distribution.png", dpi=150)
24     plt.close(fig)
```

Result



Inference

The left histogram confirms the NumPy result: most people are charged under \$15,000 with a long right tail reaching \$64,000, and the mean line sits well to the right of the median. The log-scale histogram is much more symmetric and shows **several peaks**, a hint that the data contains distinct customer groups.

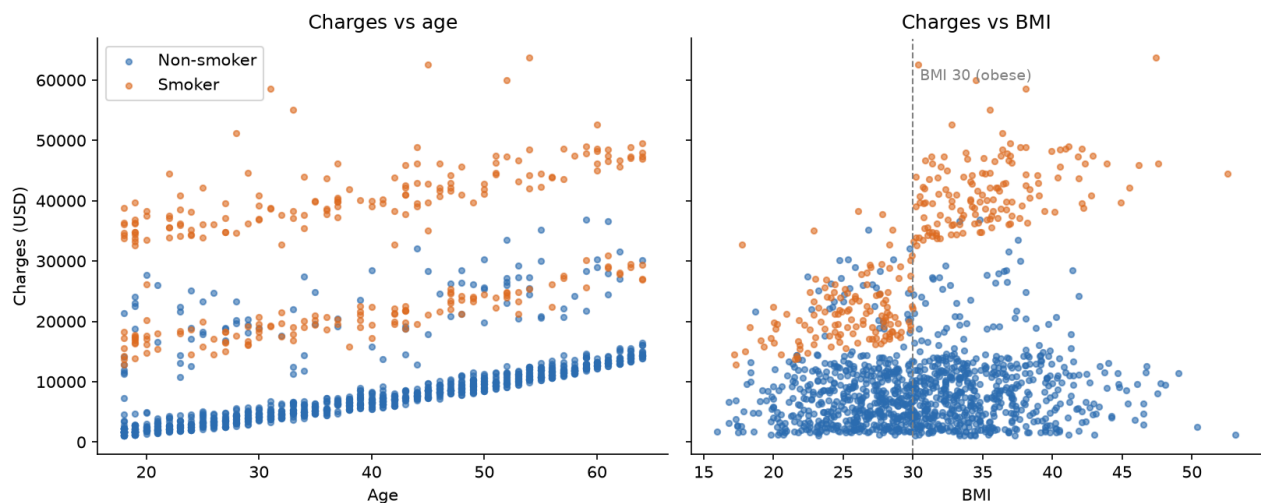
Matplotlib Application 2 - Scatter plots by smoking status

Objective: Show how charges change with age and BMI and how smokers differ.

Code

```
1 def matplotlib_2_scatter(df):
2     """Scatter plots of charges against age and BMI, coloured by smoking status."""
3     fig, axes = plt.subplots(1, 2, figsize=(11, 4.5), sharey=True)
4     for ax, col in zip(axes, ["age", "bmi"]):
5         for flag, color, name in [("no", NON_SMOKER, "Non-smoker"),
6                                   ("yes", SMOKER, "Smoker")]:
7             part = df[df["smoker"] == flag]
8             ax.scatter(part[col], part["charges"], s=12, alpha=0.6,
9                       color=color, label=name)
10            ax.set_xlabel(col.upper() if col == "bmi" else col.title())
11            ax.set_title(f"Charges vs {col.upper()} if col == 'bmi' else col}")
12            ax.spines[["top", "right"]].set_visible(False)
13            axes[1].axvline(30, color="grey", ls="--", lw=1)
14            axes[1].text(30.5, 60000, "BMI 30 (obese)", color="grey", fontsize=9)
15            axes[0].set_ylabel("Charges (USD)")
16            axes[0].legend()
17
18 fig.tight_layout()
19 fig.savefig(FIG_DIR / "mpl_2_scatter.png", dpi=150)
20 plt.close(fig)
```

Result



Inference

The age plot shows **three parallel bands**: a low band of non-smokers, a middle band mixing both groups and a top band of smokers only. Within every band charges rise linearly with age. The BMI plot reveals a clear break at **BMI 30**: smokers above it jump to \$35,000-\$50,000, while for non-smokers BMI has almost no effect. This confirms the smoker-obesity interaction visually.

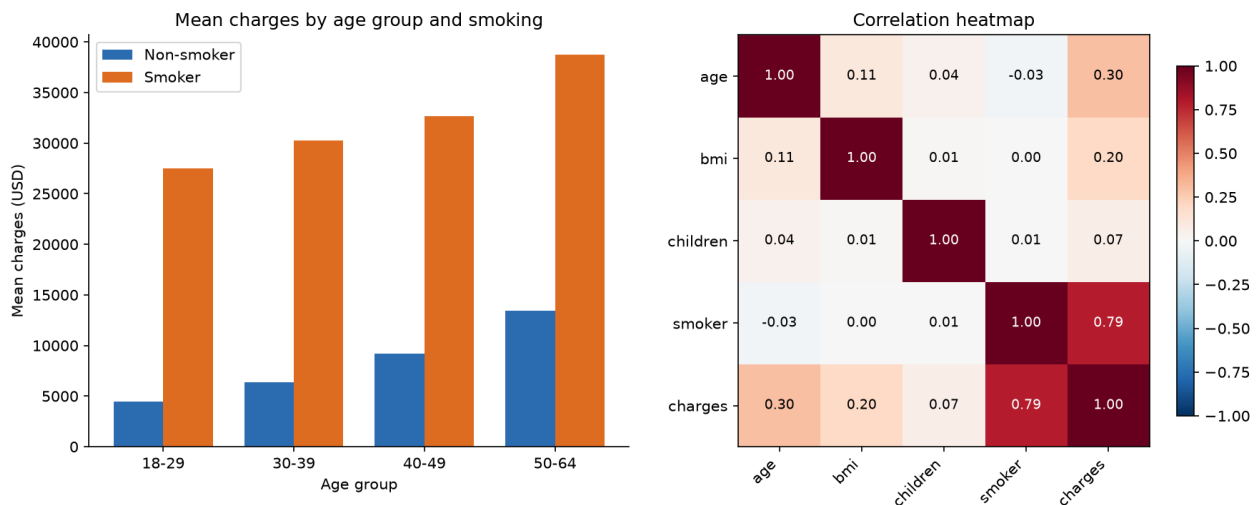
Matplotlib Application 3 - Grouped bar chart and correlation heatmap

Objective: Compare mean charges by age group and smoking status, and display the NumPy correlation matrix as a heatmap.

Code

```
1 def matplotlib_3_groups_and_heatmap(df, cols, corr):
2     """Grouped bar chart by age group x smoker and a correlation heatmap."""
3     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4.8))
4
5     table = df.pivot_table(values="charges", index="age_group",
6                             columns="smoker", aggfunc="mean", observed=True)
7     x = np.arange(len(table.index))
8     w = 0.38
9     ax1.bar(x - w / 2, table["no"], w, color=NON SMOKER, label="Non-smoker")
10    ax1.bar(x + w / 2, table["yes"], w, color=SMOKER, label="Smoker")
11    ax1.set_xticks(x, table.index)
12    ax1.set_xlabel("Age group")
13    ax1.set_ylabel("Mean charges (USD)")
14    ax1.set_title("Mean charges by age group and smoking")
15    ax1.spines[["top", "right"]].set_visible(False)
16    ax1.legend()
17
18    im = ax2.imshow(corr, cmap="RdBu_r", vmin=-1, vmax=1)
19    ax2.set_xticks(range(len(cols)), cols, rotation=45, ha="right")
20    ax2.set_yticks(range(len(cols)), cols)
21    for i in range(len(cols)):
22        for j in range(len(cols)):
23            ax2.text(j, i, f"{corr[i, j]:.2f}", ha="center", va="center",
24                    fontsize=9, color="white" if abs(corr[i, j]) > 0.6 else "black")
25    fig.colorbar(im, ax=ax2, shrink=0.85)
26    ax2.set_title("Correlation heatmap")
27
28    fig.tight_layout()
29    fig.savefig(FIG_DIR / "mpl_3_groups_heatmap.png", dpi=150)
30    plt.close(fig)
```

Result



Inference

In every age group smokers pay far more; the gap is widest in relative terms for the young (18-29: about 6x) and largest in absolute terms for the old (50-64: about \$25,000). The heatmap summarises the whole story in one picture: the only dark cell linked to charges is **smoker (0.79)**, followed by age and BMI.

Chapter 3: Conclusion

3.1 Key findings

- **Smoking is the single biggest cost driver.** Smokers are 20.5% of beneficiaries but are charged about 3.8 times more on average, account for almost all extreme-cost cases (97.8% of outliers), and add about \$23,800 to the expected charge.
- **Age raises costs steadily.** Each extra year adds roughly \$258; people aged 50-64 cost about three times as much as those aged 18-29.
- **BMI matters mainly for smokers.** Obesity nearly doubles a smoker's cost but barely changes a non-smoker's, a clear interaction effect.
- **Sex, region and number of children have little effect** once smoking is considered; the southeast's higher average reflects more smokers and higher BMI there.
- **A simple model works well.** A NumPy least-squares model with four variables explains 75% of the variation in charges.

3.2 The story in the data

The customers form three groups: a large low-cost group of non-smokers whose charges grow gently with age; a middle group of non-obese smokers (and some non-smokers with other health issues); and a small but very expensive group of obese smokers. Pandas showed the group differences, NumPy measured their size and statistical strength, and Matplotlib made the three-band pattern and the BMI-30 threshold easy to see.

3.3 Recommendations

- Price policies on smoking status, age and BMI together, with an extra loading for the smoker + BMI 30 or above combination.
- Fund smoking-cessation and weight-management programmes: moving one obese smoker to non-smoker could save about \$20,000-\$30,000 a year.
- Do not base pricing on sex or region; they add little beyond smoking.
- Add a smoker x obesity interaction term (or use a log-scale model) to improve on the R^2 of 0.75.

3.4 Limitations

The dataset is simulated from census statistics, covers only 1,338 people and lacks medical history, income and plan type, so the findings describe associations, not causes, and should be checked on real claims data.